

Ultrafast Switching of SiC MOSFETs for High-Voltage Pulsed-Power Circuits

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Abstract—Due to their fast switching capabilities, silicon carbide (SiC) MOSFETs are increasingly used in high-voltage pulsed-power circuits where fast and flexible high-voltage pulses are required (e.g. for plasma-processing applications), such as in our solid-state Impedance-matched Marx generator. The turn-on time of SiC MOSFETs mainly depends on the gate-driving technique and its implementation. Therefore, many studies focus on SiC MOSFET gate-driving methods. In this article, a gate-boosting driving method that was previously proposed for IGBTs is optimized for SiC MOSFETs and applied to a very fast MOSFET to reduce turn-on time as much as possible. A prototype of the optimized gate-boosting driver was built and tested to assess its performance. The test results validate the effectiveness of the optimized gate-driving method and showed that a MOSFET turn-on time of below 2 ns is achievable at a high operating voltage and moderate current and below 3 ns for a wider range of load-current and operating-voltage conditions. Furthermore, current rise rates of 38.7 kA/ μ s at 356-A load current are possible. All these results are considerably faster than ever demonstrated before.

Index Terms—Gate-boosting method, high-voltage pulsed-power supply, Silicon Carbide (SiC) MOSFET.

I. INTRODUCTION

FAST rise time, nanosecond high-voltage pulses are used in many applications, such as transient plasma generation for industrial and environmental applications, e.g., air purification, water purification, biological tissue sterilization, material surfaces modification, and many more [1], [2]. Here, researchers note that the short pulse duration and fast rise time of the pulses significantly increase the efficiency of a number of applications [3]–[7]. Conventionally, these fast high-voltage pulses are generated with circuits, such as the single-line pulse forming line pulse source [8], [9], Blumlein pulse source [7], [10], magnetic pulse compression pulse source [11], [12], fast diode opening switch pulse source [13]–[15], and others (a comprehensive overview can be found in [16]). While these pulse sources all deliver fast high-voltage pulses, they lack pulse waveform control (amplitude, duration, and/or shape) and/or pulse repetition

rate to make them candidates for fast, flexible high-voltage waveform generation. To overcome this, many researchers turn to pulse sources utilizing a series/parallel connection of fast semiconductor switches (i.e. MOSFETs and IGBTs), such as the solid-state Marx generator [17]–[19] or the solid-state Linear Transformer Driver [20]. Recently, we introduced the solid-state Impedance-matched Marx generator (IMG) as an even faster alternative [21]. However, to maximize the potential of such a topology, we need to significantly increase the switching speed of standard semiconductor devices, which is the topic of this article.

Since the introduction of wide bandgap semiconductor devices, their rated voltage and current continue to increase with every generation. This is especially true for silicon carbide (SiC) devices, which makes them a good choice for pulsed power circuits [22], [23]. For fast switching, the SiC MOSFET is the best choice. Since the switching speed of such MOSFETs is directly affected by the method to drive the gate of these devices [23], many studies are concentrated on finding proper driving techniques for fast switching applications.

Before we move on to other studies of fast gate-driving methods, we need to consider the switching requirements for our application: the solid-state IMG [21]. The first consideration is the complexity of the gate-drive circuit. Since we will have to use a great number of SiC MOSFET switches in the solid-state IMG (in parallel and in series), the control complexity of the gate-drive circuit should not be excessive. Especially, the number of control signals to the drive circuitry should be as low as possible because these all need to be isolated, either optically or through other means. Second, for the same reason (using a large number of switches), the cost of the gate-drive circuit should not be excessive. Finally—and perhaps most importantly—the gate-drive circuit should drive the MOSFETs at the fastest speeds possible to ensure fast-rising high-voltage pulses from the solid-state IMG. Our aim is a voltage rise time¹ of just several nanoseconds.

The turn-on time of SiC MOSFETs directly depends on the gate current (and the gate–source capacitance and parasitic elements). Therefore, any improvement in switching time requires increasing the gate current. The peak gate current is limited by the gate drive loop’s resistance and inductance (internal or external) and also the driving voltage amplitude.

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¹Since the switches are compactly connected to transmission lines in the solid-state IMG, the current rise time will be very similar to the voltage rise time.

Therefore, in addition to minimizing gate loop parasitic impedance, any increment in gate current needs increasing the gate drive voltage level.

Some solutions for reducing the semiconductor's turn-on time are proposed in the literature. These methods are based on applying a high voltage to the gate (higher than its rated voltage) for a very short time. For instance, in [24]–[27], current-source gate drive circuits based on a full-bridge topology are suggested to improve the switching speed of MOSFETs. Although these methods will decrease the switching time, they need extra driving switches and control circuitry, which increases the complexity, cost, and size of the gate driver circuit. Furthermore, other types of gate drivers based on applying variable voltage and current to the SiC MOSFETs are proposed in [28] and [29]. Although these drivers are effective on the MOSFET's turn-on time (by almost a factor of two in both studies), they need more switches and a complicated control mechanism. In addition, an interesting approach to fast gate driving of a SiC MOSFET is proposed in [30]. Here, they focus on maximizing the current rise rate of the switch and make use of inductive storage to supply a current to the gate, which uses inductive kickback to generate a transient overvoltage on the gate. Although this technique does not need a significant amount of components, it requires a complicated switch control system.

Perhaps, the most interesting gate drive for our purpose is the work presented by Hochberg *et al.* [31], [32]. Here, a gate-boosting driver is proposed and investigated for SiC IGBTs (and also tested on a SiC MOSFET) with very promising results. The tested IGBTs and MOSFETs increased their current rise time by a factor of 8 and 3.5, respectively, when using the gate-boosting circuit. Furthermore, this gate-boosting method does not require extra switches and a sophisticated control mechanism in comparison with the other fast gate drivers in the literature. Hence, it can be used in a modular high-voltage pulsed power supply without introducing a high complexity to the system. To push SiC MOSFET switches to their switching limit for our application in the solid-state IMG, we will modify the gate-boosting circuit proposed by Hochberg *et al.* for even better performance.

This article is arranged in six sections. After this introduction, Section II introduces the optimized gate-boosting driver for SiC MOSFETs, followed by extensive simulations of this drive in Section III. Then, in Section IV, the implementation of the optimized driver and its test method is described. Subsequently, the experimental results are presented and discussed in Section V. Then, Section VI presents the conclusions.

II. OPTIMIZED GATE-BOOSTING CIRCUIT FOR SiC MOSFETs

The original gate-boosting circuit of [31], [32] is depicted in Fig. 1(a). It is shown here with an IGBT, but it works also with MOSFETs. This circuit works based on reducing the time constant of the RLC circuit formed by the total resistance, inductance, and capacitance in the loop of the gate drive circuit. Here, the inductance and resistance are partly internal and partly external to the MOSFET, and the capacitance is mainly

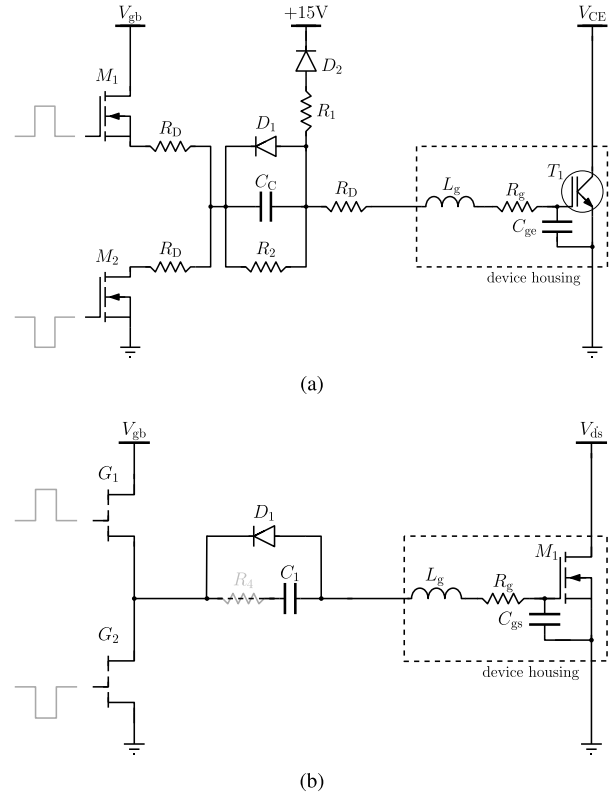


Fig. 1. (a) Conventional gate-boosting circuit as presented in [31]. (b) Optimized gate-boosting circuit for fast switching of SiC MOSFETs. Resistor R_4 is only shown for the discussion on the performance of the optimized circuit; it is not used in the final implementation.

formed by the gate–source capacitance. By adding coupling capacitor C_C in the gate-drive line, the total capacitance of the RLC loop decreases, resulting in a lower time constant and faster charging of the gate. At the same time, the circuit applies a transient overvoltage to the gate–source terminals for even faster switching. The MOSFET gate survives this because the SiC MOSFET die in the MOSFET package does not experience this transient overvoltage (due to the internal inductance) [31]. The final value of the gate voltage is determined by the capacitive divider formed by C_C and the gate–source capacitance of the MOSFET and should not exceed the rated value.

We optimized the conventional gate-boosting circuit in the following way (see Fig. 1(b)).

- 1) The current limiting resistors, R_D , are removed to increase the gate charge and discharge current to the highest possible value. This instantaneous high current peak should be considered in selecting the driver's switches. In Fig. 1(b), R_4 is only shown for the discussions. It is not used in the final implementation.
- 2) The voltage protection circuit (R_1 , D_2 , and +15 V) is eliminated. As mentioned earlier, the gate-boosting circuit works based on applying a high voltage to the MOSFET's gate. Therefore, keeping the gate voltage in a determined value by the protection circuit reduces the effectiveness of the gate-boosting circuit. The gate-boosting voltage V_{gb} is calculated with the internal gate impedance to assure MOSFET safety.

By correctly selecting the gate-boosting voltage and capacitor, the MOSFET's gate on the die does not experience a voltage higher than its rated value. We will discuss this in the simulation result, and it is also described in [30]–[32].

- 3) The resistor R_2 of the conventional gate boosting circuit is removed in the optimized version. In the optimized version, the gate-boosting capacitor is selected to keep the gate steady-state voltage below the maximum allowed transient value. Therefore, even in long pulse applications (microseconds' level), it always has an acceptable value. Hence, the optimized circuit does not need R_2 to keep the steady-state gate voltage above an acceptable value. For very high repetition rate (above tens of kHz) operation one might consider to read R_2 for a faster discharge of C_1 .
- 4) The driver's switches are replaced with fast Gallium Nitride (GaN) MOSFETs to improve the gate driver's speed.

The operation principle of the optimized gate-boosting circuit can be explained in two sections. During turn-on, the driver's upper switch turns on, and the gate-boosting voltage is applied to the MOSFET gate through R_4 – C_1 . First, when C_1 is almost discharged, the main gate current passes through R_4 – C_1 . Hence, R_4 , C_1 , and the gate-boosting voltage are effective parameters to influence the MOSFET turn-on behavior. Although C_1 has a considerable effect on the MOSFET switching speed, its value is determined by the rated gate voltage. The value of C_1 determines the final voltage on the MOSFET's gate and can be calculated with (1). In this equation, V_{gb} is the gate-boosting voltage, V_{gr} is the rated gate voltage, and C_{iss} is the input capacitance of the MOSFET (at the drain–source voltage operating point). Since the final application of the proposed gate driver is the pulsed power supply and the pulse duration is much shorter than the pulse repetition time, the transient value of gate rated voltage can be used in (1), which decreases the output pulse's rise time

$$C_1 = \frac{V_{gr} C_{iss}}{V_{gb} - V_{gr}}. \quad (1)$$

To turn off the MOSFET, the lower switch of the bridge turns on, and the MOSFET input capacitance starts to discharge via R_4 – C_1 . In this state, C_1 works as a negative voltage source, which aids the fast turn off of the SiC MOSFET in a high-noise environment. When the voltage of C_1 reaches zero and it begins to charge in the reverse direction, the MOSFET's gate–source capacitance (C_{gs}) continues to discharge through D_1 .

III. SIMULATION OF OPTIMIZED GATE-BOOSTING CIRCUIT

The optimized gate-boosting circuit is simulated with LTspice [33] to evaluate the impact of the component values on its performance and also to evaluate the switching behavior of various types of SiC MOSFETs. In the simulation, two GaN MOSFETs (GS61004B) are used as driver switches. Fig. 2 shows the circuit used for these simulations.

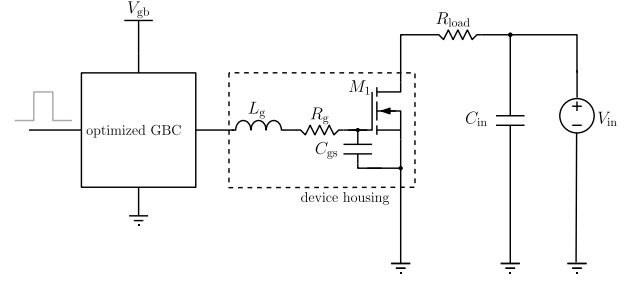


Fig. 2. Circuit used for the simulations. The block “optimized GBC” represents the circuit of Fig. 1(b).

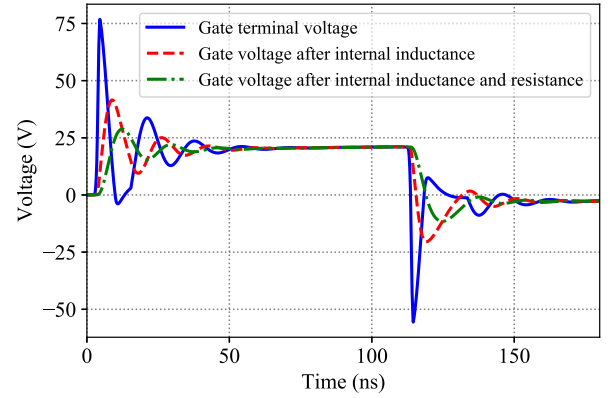


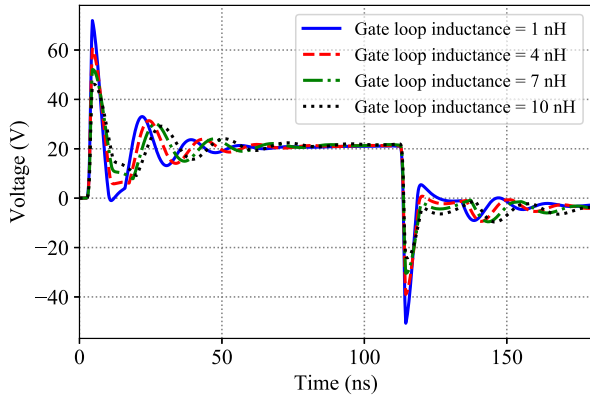
Fig. 3. Simulated SiC MOSFET gate path voltage, illustrating the different voltages present in the MOSFET.

A. Evaluation of the Effect of Circuit Component Values in the Optimized Gate-Boosting Circuit

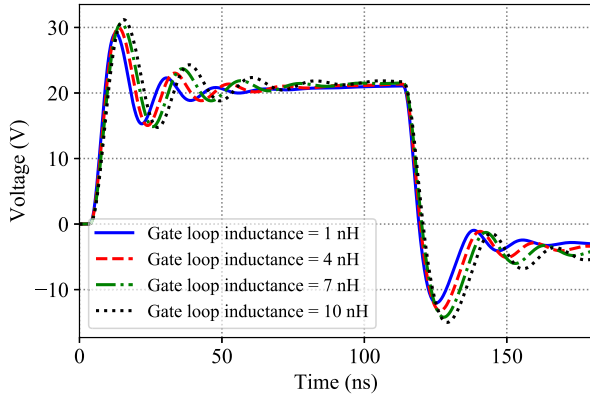
The optimized gate-boosting circuit is simulated in various conditions to separately illustrate the impact of any component value on its performance. In the first section, the effect of applying a voltage higher than the gate rated voltage is investigated. The MOSFET is replaced with an RLC circuit that models its gate circuit. The effect of the Miller capacitance (C_{gd}) is neglected. Therefore, only C_{gs} is used in the simulation. The values of the MOSFET's parameters (R_g , L_g , and C_{gs}) are taken from the official LTspice model. As the simulation result shows (see Fig. 3), although the voltage of the MOSFET's terminal rises more than its rated voltage, the die does not experience this high voltage, and its voltage does not exceed the rated voltage.

The effect of the gate loop's parasitic inductance is depicted in Fig. 4. Based on the results, increasing the gate loop's inductance reduces the voltage peak and the oscillation at the MOSFET's gate terminal. Conversely, the increase in inductance increases the voltage oscillation at the die and increases the rise time, reducing switching speed. The lead's inductance has a similar effect. Therefore, the gate-driving circuit should be designed with as little inductance as possible to optimize the voltage at the die and, thereby, optimize switching speed.

Next, the impact of the driver supply voltage (V_{gb}) is studied, by replacing the RLC model of the gate with the actual MOSFET model (in our case, the C2M0045170D



(a)

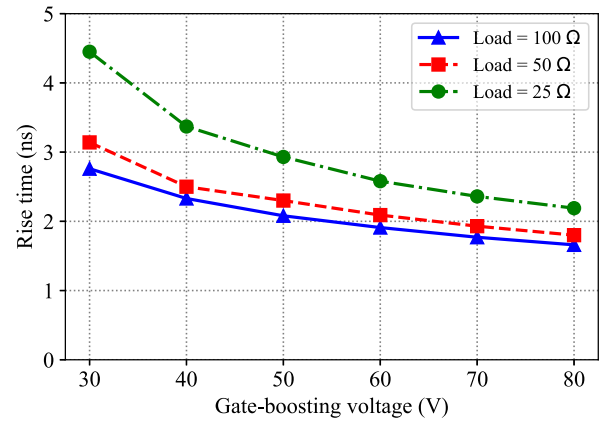


(b)

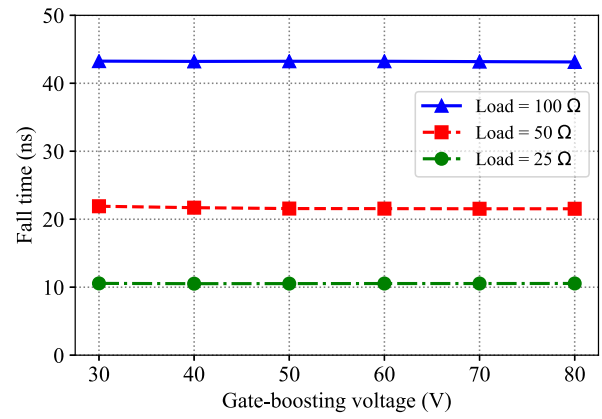
Fig. 4. Simulated effect of the gate loop's parasitic inductance on the gate voltage for (a) gate terminal voltage and (b) die voltage.

model, a SiC MOSFET from Wolfspeed). Fig. 5 shows the variation of the current rise and fall time versus gate-boasting voltage (here, we assume no inductance in the load for now). An appropriately-sized proper gate-boasting capacitor is calculated with (1) and used in each step of gate-boasting voltage. Increasing the gate-boasting voltage increases the gate charging current and, therefore, reduces the rise time, as can be seen in the results. On the other hand, it does not have a significant effect on the fall time. The turn-off behavior of the MOSFET, especially in low-load conditions, mainly depends on the load current, and the gate drive circuit almost has no control over it. Therefore, in the remainder of this article, the focus will be on the turn-on behavior of the MOSFET.

The other optional component in the gate-boasting circuit to analyze is R_4 . This resistor, which is in series with the MOSFET turn-on path, limits the charging current of C_{iss} and, therefore, has a direct effect on the rise time. The impact of R_4 on the MOSFET turn off is smaller than its role on turn on. However, it limits the MOSFET's gate discharge current and is, therefore, effective on the SiC MOSFET's fall time and turn-off delay. Fig. 6 shows the effect of R_4 on the gate-boasting circuit's performance. As can be seen, the best performance of the gate-boasting circuit is achieved when R_4 is zero. Therefore, R_4 is removed in the optimized version of the gate-boasting circuit.



(a)



(b)

Fig. 5. Simulated impact of the gate-boasting voltage on MOSFET switching time, with V_{in} (input voltage) = 1200 V. (a) Current rise time and (b) current fall time.

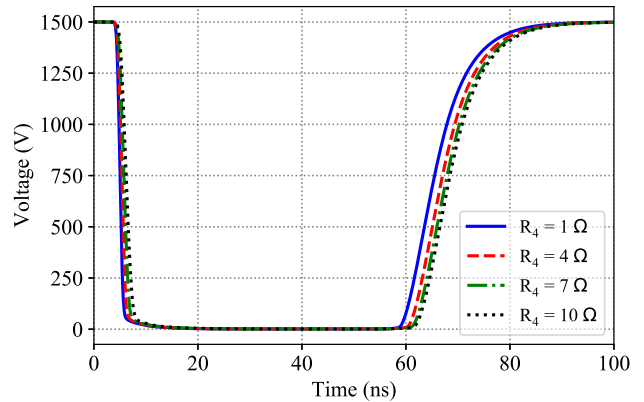


Fig. 6. Simulated effect of R_4 on SiC MOSFET switching times.

B. Evaluation of Various SiC MOSFETs With the Optimized Gate-Boasting Driver

After the above investigation on the optimized gate-boasting circuit, in this section the switching performance of some SiC MOSFETs is studied using this driving circuit. Table I shows the part number and parameters of the tested MOSFETs, where V_{ds-r} and I_d are the rated value of the SiC MOSFET's drain-source voltage and drain current, respectively, V_g is the rated

TABLE I
COMPARED SiC MOSFET PARAMETERS

Part No	V_{ds-r} (V)	I_d (A)	V_g (V)	C_{iss} (nF)	C_{rss} (pF)	C_{oss} (pF)	R_g (Ω)	L_g (nH)
C2M0045170D	1700	72	20	3,67	6,7	171	1,3	8,47
C2M0045170P	1700	72	20	3,67	6,7	171	1,3	7,33
C2M0080170P	1700	40	20	2,25	4	105	2	12,42
C3M0075120J	1200	30	15	1,347	2	58	9	7,79
C2M0040120D	1200	60	20	1,893	10	150	1.8	15

TABLE II
CALCULATED TURN-ON TIME OF THE SELECTED SiC MOSFETs

Part No	turn-on time (ns)
C2M0045170D	2,51
C2M0045170P	2,51
C2M0080170P	1,83
C3M0075120J	6
C2M0040120D	2,7

gate–source voltage of the MOSFET, C_{iss} , C_{oss} , and C_{rss} are the MOSFET's input, output and reverse transfer capacitance, respectively, and R_g and L_g are the MOSFET's internal gate resistance and inductance, respectively.

First, the turn-on time of these SiC MOSFETs is estimated using a theoretical analysis. We used the estimation method proposed in [23], which considers the nonflat plateau region of SiC MOSFETs and seems more accurate than other methods. The switch turn-on time (drain–source voltage fall time) t_{ON} is calculated using (the gate path inductances are neglected in this method)

$$t_{ON} = R_g \left(C_{iss} \ln \frac{V_{dr} - V_{gp1}}{V_{dr} - V_{gp2}} + \frac{C_{rss}}{K_1} \ln \frac{V_{dr} - K_2 - K_1 V_{ds-ON}}{V_{dr} - K_2 - K_1 V_{ds-max}} \right) \quad (2)$$

where V_{dr} is the gate-driving voltage, V_{ds-ON} and V_{ds-max} are the ON-state and OFF-state drain–source voltages of the MOSFET, respectively, V_{gp1} and V_{gp2} are the start and the end of the gate–source voltage plateau region, respectively (calculated with the datasheet), and K_1 and K_2 are defined as

$$K_1 = \frac{V_{gp2} - V_{gp1}}{V_{ds-ON} - V_{ds-max}}, \quad K_2 = \frac{V_{gp1} V_{ds-ON} - V_{gp2} V_{ds-max}}{V_{ds-ON} - V_{ds-max}}. \quad (3)$$

The turn-on time of the MOSFETs mentioned in Table I is calculated using (2) under rated gate–source and drain–source voltage conditions. The calculation results are tabulated in Table II. The results show that C2M0080170P is the fastest MOSFET and could be used in our IMG.

In the second step, the SiC MOSFET's performance is simulated with the optimized gate-boosting driver (with the same simulation circuit of Fig. 2) to find the appropriate switch for the IMG and also to assess the calculations. Fig. 7 shows the current rise time of the selected SiC MOSFETs when the input voltage (V_{in}) is 1200 V and the load is 100 Ω and 25 Ω . In each case, the simulation is started with the gate rated voltage and conventional driving method (the first point in the figure). For higher voltages, the gate-boosting circuit

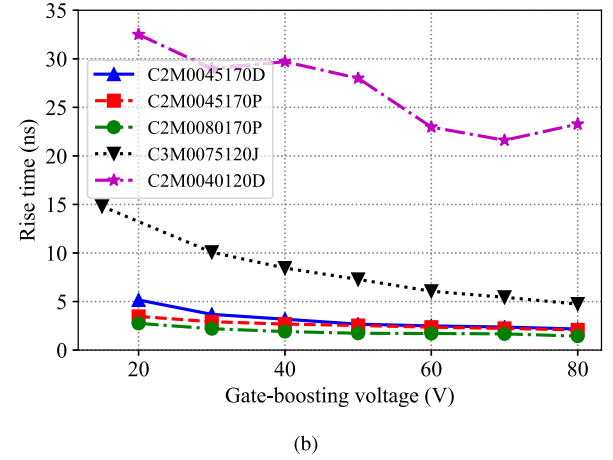
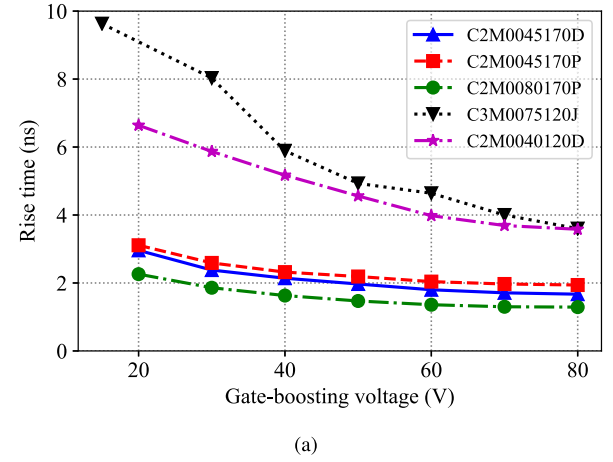


Fig. 7. Simulated SiC MOSFET current rise time for the selected MOSFETs, with $V_{in} = 1200$ V and (a) $R_{load} = 100 \Omega$ and (b) $R_{load} = 25 \Omega$.

is used. Simulations are done with a proper value of the gate-boosting capacitor calculated using (1), which guarantees that the die voltage always remains under its permitted value, with a steady-state value equal to the gate's rated transient voltage. As expected, the rise time reduces when the gate driver voltage increases. The amount of the decrease depends on the gate circuit's internal component values. The greatest variations in rise time with the gate-boosting voltage occur with the MOSFET with the highest internal gate resistance and inductance value (the C3M0075120J and C2M0040120D in our selection).

As expected, the rise time increases when the load current increases (see Fig. 7(b)). However, increasing the gate-boosting voltage in this situation is still very effective. The

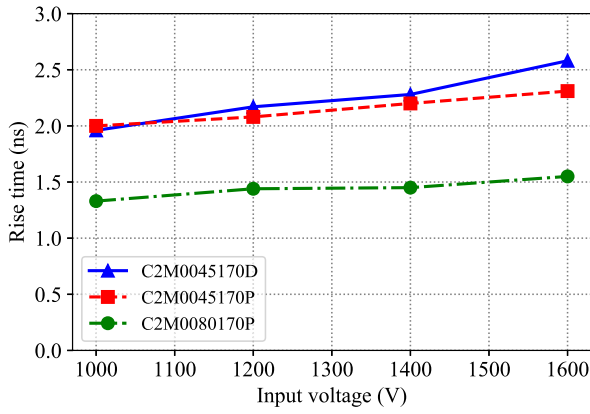


Fig. 8. Simulated SiC MOSFET current rise time as a function of the drain–source voltage, with $V_{gb} = 80$ V and $R_{load} = 25$ Ω .

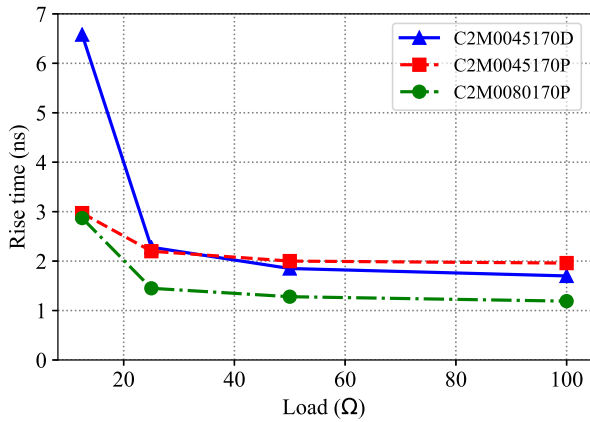


Fig. 9. Simulated SiC MOSFET current rise time as a function of the load resistance, with $V_{gb} = 80$ V and $V_{in} = 1400$ V.

results demonstrate that the gate-boosting circuit performance in all cases is better than the conventional driving method. It should be mentioned, with a resistive load, that the current rise time and V_{ds} fall time have the same value. In reality, any difference between the voltage fall time and the current rise time is imposed by the inductance of the load.

In addition, some simulations are done to investigate the effect of the input voltage (V_{in}) on MOSFET switching behavior. Fig. 8 shows the results of the simulations. Since increasing the input voltage and load current increases the required turn-on charge of the gate, the rise time of MOSFET will increase.

Finally, the effect of the load resistance variation on MOSFET switching performance is investigated. During the simulations, V_{in} was 1400 V, and the load is increased from 12.2 to 100 Ω . Fig. 9 presents the results, which shows that, as the load current decreases, the rise time reduces.

Both calculation and simulation results show that the C2M0080170P MOSFET has the best switching performance and, therefore, is the best choice for use in our IMG. Finally, according to the simulations, using the modified gate-boosting circuit in ideal conditions, the C2M0080170P can achieve less than 2-ns rise time in all loading conditions and under different drain–source voltages. Also, this device has a built-in Kelvin



Fig. 10. Implemented circuit schematic.

source connection pin that is very useful to reduce the coupling between the gate-driving circuit and the high-power circuit. Therefore, it is selected to study its performance in the various experimental tests presented in Section V.

IV. IMPLEMENTATION OF THE MODIFIED GATE-BOOSTING CIRCUIT

A test circuit is designed and implemented based on the schematic in Fig. 2. To test the performance of the C2M0080170P MOSFET in conditions as close as possible to when it would be used in the real IMG, a fiber optic link is used to transfer the pulse command to the test board. The received command by the fiber optic receiver is amplified and fed to a half-bridge driver that controls the gate-boosting circuit. The gate-boosting circuit, in turn, switches the MOSFET. The implemented circuit schematic is shown in Fig. 10.

Fig. 11 shows the printed circuit board (PCB) that we have designed for the optimized gate-boosting driver testing. The PCB is designed to have as short as possible power and gate loops, and also parallel paths are used for the high current path to reduce the parasitic inductance. In addition, because of shoot through in the driver's switches, a sufficient amount of capacitance is placed as close as possible to the driver's switches on the top and bottom side of PCB to buffer the gate-boosting voltage. Two GaN MOSFETs (GS61004B) are used as driver switches. The main SiC switch is the C2M0080170P MOSFET that had the best performance in the calculation and simulation results. The value of the gate-boosting capacitor is chosen for each experiment to ensure that the die voltage always remains under its permitted value. To supply the output current pulse, a sufficient amount of capacitors of the type C0G and X7R is placed on a high-voltage bus on both sides of the PCB. Also, the load is constructed by paralleling resistors to reduce the high-power path inductance, which reduces the difference between drain–source voltage fall time and load current rise time.

The output current is measured by a homemade Rogowski coil that can be fitted around a MOSFET pin. The details on this sensor can be found in the Appendix.

V. EXPERIMENTAL TEST RESULTS

In this section, the presented optimized gate-boosting driver is tested to evaluate its performance under various conditions. Fig. 12 shows the drain–source voltage and the load current at different input voltages and load current levels. Fig. 13 depicts the gate–source voltage of the SiC MOSFET. As can be seen, the gate–source voltage has a high peak value during turn-on and a negative peak during turn-off, both of which are not experienced by the device's gate on the die. All high voltages in the tests are measured with a Lecroy high-voltage probe (PPE6KV) with 400-MHz bandwidth.

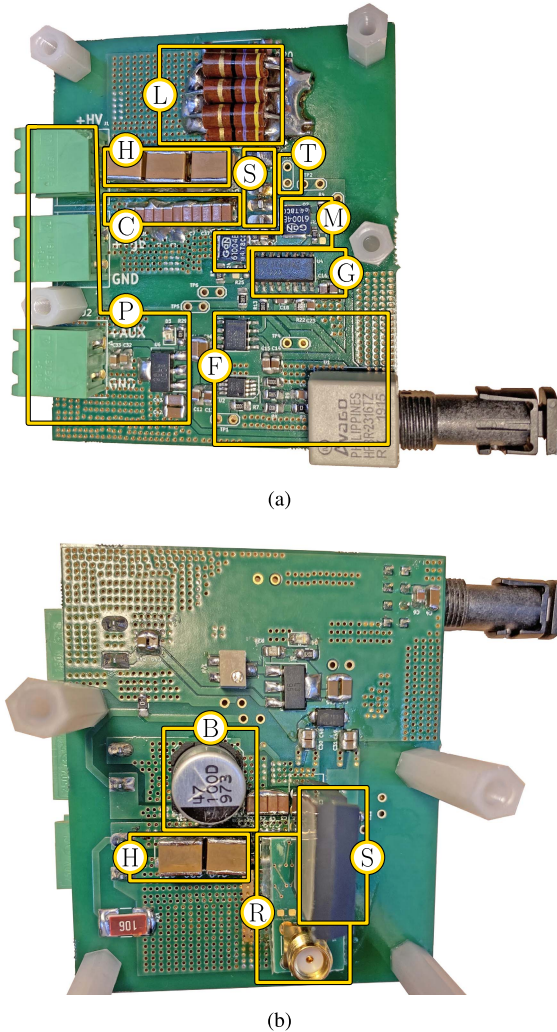
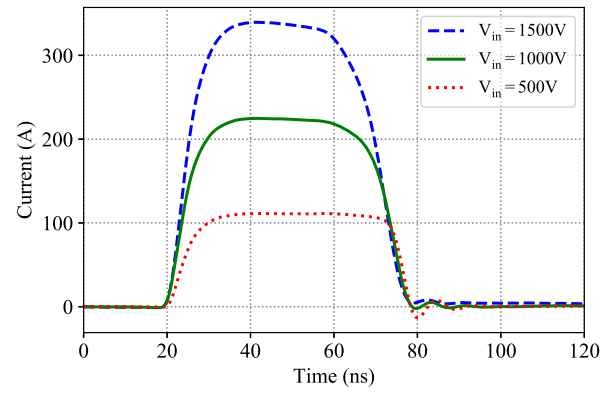
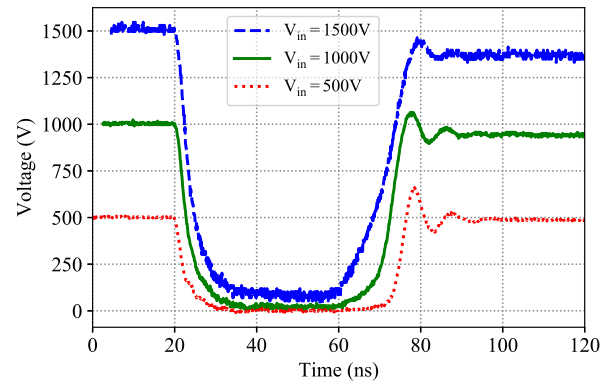


Fig. 11. Designed PCB for optimized gated-boosting driver test. (a) Top side. (b) Bottom side. The components on the PCB are load resistors (L), HV capacitors (H), gate-boosting voltage's buffer capacitors (B), SiC MOSFET (S), modified gate-boosting circuit (M), the GaN's half-bridge driver (G), power supply input and conditioning (P), fiber optic receiver and interface circuit (F), the Rogowski coil (R), and the voltage probe test point (T).

To demonstrate the effect of the gate-boosting voltage on the MOSFET's switching speed, the output current's rise time and V_{ds} fall time versus gate-boosting voltage for various load conditions are measured and shown in Fig. 14. In addition, we also built an identical test PCB with regular fast silicium (Si) MOSFETs (SIA446DJ) in the gate-boosting driver's totem pole (instead of the GaN switches) and compared that to the version with the GaN switches. The result of this comparison is also shown in Fig. 14. The test started with conventional gate driving (rated gate voltage without the gate boosting capacitor), and at each next step in increasing the gate boosting voltage, the proper capacitor is added to the circuit. The rise and fall times are calculated by averaging over 200 samples. As expected, increasing the gate-boosting voltage decreases the turn-on time of the SiC MOSFET. The results also show that the gate-boosting circuit with GaN driver switches outperforms the circuit with Si switches, showing that the use of the GaN switches is a valuable addition to the optimized gate-boosting circuit.



(a)



(b)

Fig. 12. Some examples, measured output pulses at different input voltages, with $V_{gb} = 95$ V and $R_{load} = 4.5$ Ω . (a) Drain current and (b) drain-source voltage.

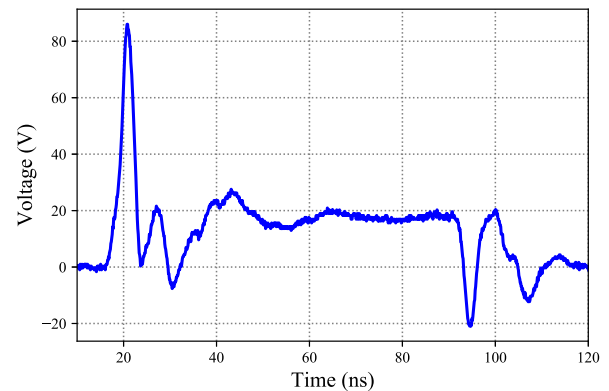
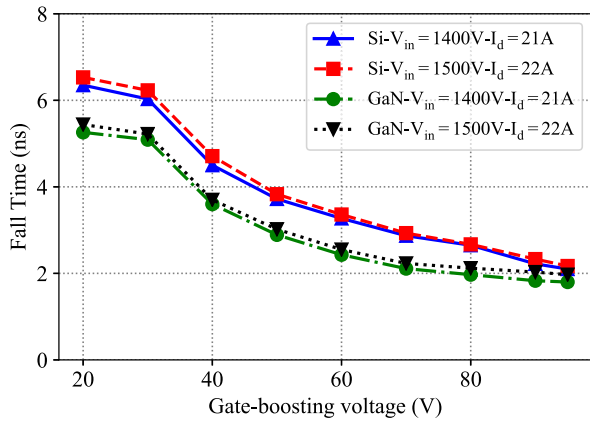


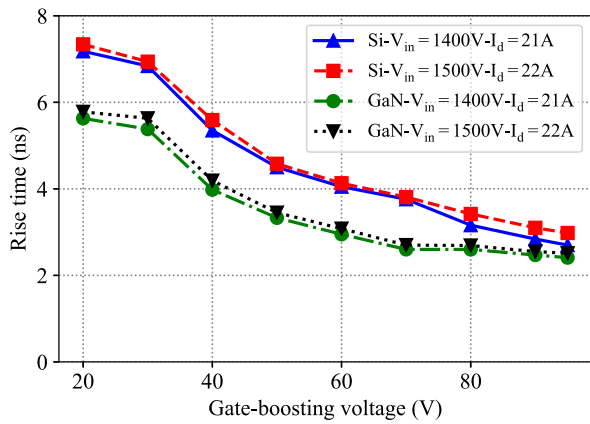
Fig. 13. SiC MOSFET gate-source voltage with the gate-boosting circuit.

The results further demonstrate that, by using the optimized gate-boosting circuit, a 2-ns voltage fall time and 2.5-ns current rise time are achievable with 1500-V input voltage at around a 20-A load current. As mentioned in Section III, the difference between the voltage fall time and the current rise time is the result of inductance in the load path.

Another test is done to investigate the performance of the gate-boosting circuit for various load currents and input voltages. The test results are shown in Fig. 15. As can be seen, with constant V_{in} , a higher load current increases the



(a)



(b)

Fig. 14. Effect of the gate-boosting voltage on the switching speed. (a) V_{ds} fall time and (b) current rise time.

turn-on time. Also, the drain–source voltage increases the turn-on time, especially in light load conditions. For higher load currents, the input voltage increment decreases the V_{ds} fall time up to 1 kV, and then, it starts to increase very slightly. This different behavior mainly originates from the nonlinear $C_{iss}/C_{rss} - V_{ds}$ curve of the SiC MOSFET. When the drain–source voltage is high enough, the pulse rise time (10%–90% of the final pulse value) completely occurs in the linear region of the $C_{iss}/C_{rss} - V_{ds}$ curve (see (2)). Therefore, the curve’s nonlinear region has no effect on the pulse rise time, and a shorter rise time is achievable [28]. As can be seen in Fig. 15, a fall time of below 3 ns is achievable for a wide range of conditions with the optimized gate-boosting driver.

In the next test, the rise time of the output pulse current and the fall time of the output pulse voltage for a constant load (4.5Ω) and various input voltages are measured, and the results are shown in Fig. 16. Based on the results, the switching speed decreases when the input voltage and drain current increase. Concluding, using the optimized gate-boosting technique achieves an output current with a rise time lower than 9 ns in a wide output current range. As mentioned in [30], to some extent, the increase in current rise time can be the result of package inductance (TO247) that is not present in the case of using a bare SiC die. Therefore, the result will

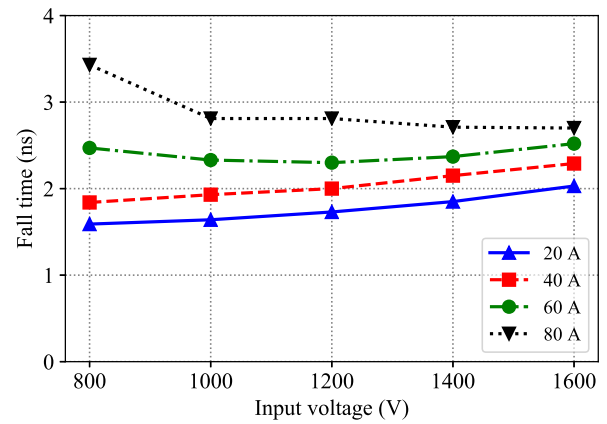


Fig. 15. V_{ds} fall time versus input voltage for different load conditions.

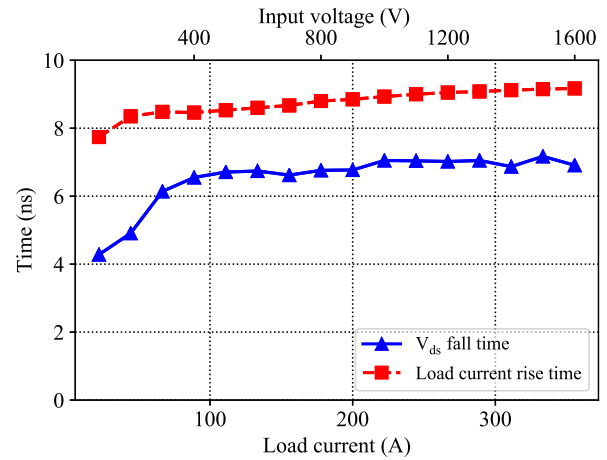


Fig. 16. Output voltage and current pulse rise time, with $V_{gb} = 95$ V and $R_{load} = 4.5 \Omega$.

likely be even better when using a bare die (which was not done in our study).

The current rise time results show that, using the optimized gate-boosting driver and with a big package MOSFET (TO247), a current rise rate (dI/dt) of $38.7 \text{ kA}/\mu\text{s}$ at 356-A load current is achieved. The presented results in [30] for a similar SiC MOSFET at the same load current are almost $13.5 \text{ kA}/\mu\text{s}$. Also, in [30], $20 \text{ kA}/\mu\text{s}$ was reached for a bare die same of the same MOSFET at the 300-A load current. This comparison demonstrates that our optimized gate-boosting driver can produce very short rise time pulses compared with other circuits. Here, it should be mentioned that, in [30], a slightly different Wolfspeed SiC MOSFET was used (the C2M0045170D, where we used the C2M0080170P). As we can see in Figs. 7 and 8 and Table II, the MOSFET that they used is slightly slower in simulations, so it stands to reason that their results could be improved when they use the same MOSFET as we do. However, even then the difference is still considerable, especially when we consider that they saw a significant switching-speed increase when they used the bare die, which is something that we have not attempted yet and could even further increase the switching speed with the optimized gate-boosting circuit.

In our final application (the solid-state IMG), the gate-boosting circuit will be used to switch MOSFETs that are connected to compact transmission lines [21], and the difference in the voltage fall time and the current rise time will be less than presented here with the resistors as a load. Furthermore, each MOSFET in the IMG will be used to switch only part of the current. Therefore, we will operate the MOSFETs more to the lower end of the load current results presented in this section. Combined with the use of transmission lines, we expect that the current rise time in our application will be much faster than the 9 ns presented here.

Finally, we assessed the MOSFET reliability when using the gate-boosting circuit. Aggressive driving of the SiC MOSFET gate could potentially cause long-term effects that limit the device's lifetime. While a much longer study would be needed to test this, we have made a first attempt by repeated switching of a MOSFET with our gate-boosting circuit. We tested the optimized gate-boosting circuit and the selected SiC MOSFET at 1.4-kV input voltage and 100-A load current for 10-M shots (350-Hz pulse repetition frequency for 8 h). The measurement results did not show any change in the gate-boosting circuit and the SiC MOSFET performance during the test.

VI. CONCLUSION

In this article, an optimized version of a gate-boosting driver circuit for SiC MOSFETs is proposed for ultrafast switching, and the role and effect of its components are discussed in detail. Based on simulations and experimental test results, the optimized driver has good performance and can be used for ultrafast switching of SiC MOSFETs in fast pulsed power applications. The results show that a MOSFET turn-on time of below 2 ns is achievable at a high operating voltage and moderate current and below 3 ns for a wider range of load-current and operating voltage conditions. Furthermore, the output current pulse's rise time for a wide range of load conditions is below 9 ns, and a current rise rate of 38.7 kA/ μ s is achieved at 356-A load current, which is significantly higher than the results of other driving techniques. In addition, the rise time could be even lower than 9 ns if the SiC MOSFET die is used, and it will also be lower in our final application (the solid-state IMG), where the load of the MOSFET circuit is a compact transmission line (so very low inductance) and we will switch in a low-current regime per MOSFET.

APPENDIX

CURRENT MEASUREMENTS WITH A HOMEMADE ROGOWSKI COIL

The output current is measured by a homemade Rogowski coil that can be fitted around a MOSFET pin and has acceptable bandwidth [34], [35]. The Rogowski coil with six turns is designed to work in the differentiating regime in a wide frequency range [36] and is implemented in a separate PCB which is placed around the SiC MOSFET's drain pin to be as compact as possible, thereby preventing adding extra inductance in the power loop (which would occur by using a regular, larger current sensor). Fig. 17 depicts the designed

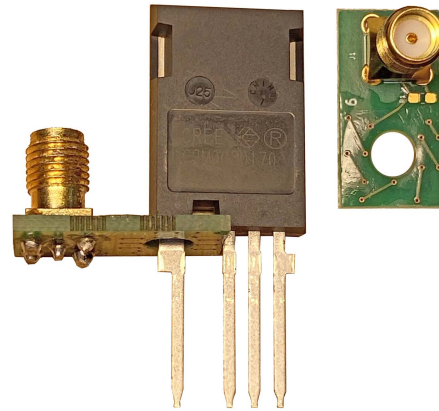


Fig. 17. Designed PCB-based six-turn Rogowski coil that can be fitted around a MOSFET pin to prevent adding extra inductance in the power loop. Left: fitted around MOSFET pin. Right: top view.

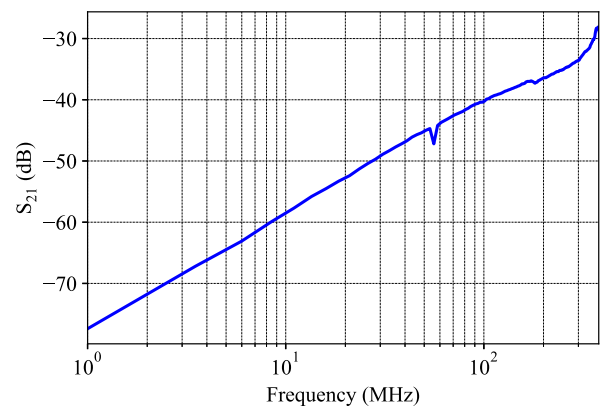


Fig. 18. Measured frequency response of the designed six-turn Rogowski coil.

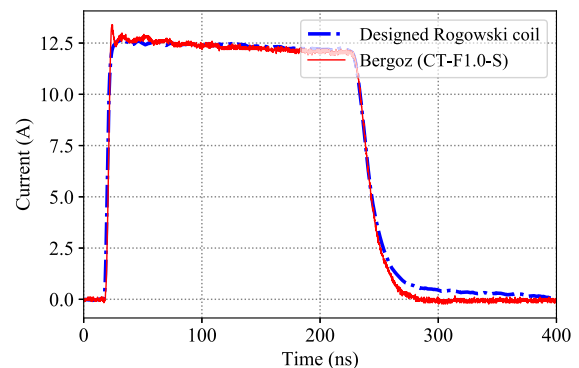


Fig. 19. Example of the measured current with the Bergoz current sensor (CT-F1.0-S) and the designed small six-turn PCB Rogowski coil that can be fitted around a MOSFET pin.

PCB-based Rogowski coil. The designed Rogowski coil has an SMA connector that can be directly connected to the oscilloscope or other data acquisition devices and passive or active integrators.

The frequency response of the designed Rogowski coil is measured by an Agilent Network Analyzer (E5061B) and depicted in Fig. 18. Based on the measured S_{21} , the Rogowski

coil works in a linear differentiating regime to almost 350 MHz, which is sufficient to measure the rise time of the current pulse. Subsequently, the current can be calculated using simple integration and applying an amplitude calibration gain.

A Bergoz current sensor (CT-F1.0-S) with a 500-MHz bandwidth is used to calibrate the Rogowski coil. Although the calibration gain varies slightly with current and coil geometry, these variations are limited in a narrow band. Therefore, it can be considered constant for many applications without imposing significant error to the calculations. Fig. 19 shows an example of the measured current with the Bergoz current sensor, and the designed Rogowski coil and the results show good agreement.

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